# NP-completeness exercises

## 34-E01 Nondeterministic paths and verifiers

Starting from the NP verifier definition, prove that polynomial-length certificates are equivalent to polynomial-time nondeterministic accepting computations. Give both directions and identify the source of soundness and completeness. This is a research-level exercise.

## 34-E02 Bit length and pseudopolynomial subset-sum

For $m \geq 2$, set $n = m, t = B = 2^m, a_i = 2^{m-1} + i, i = 1, \ldots, m$. Find the asymptotic binary input length N and prove that the standard full-table DP cost $\Theta(nt)$ is not polynomial in N. Distinguish an asymptotic family, one fixed instance, and fully unary encoding. This is an advanced exercise.

## 34-E03 Full three-CNF-SAT to clique construction

Use $F = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee \neg x_3) \wedge (x_1 \vee x_2 \vee x_3)$. Create groups of three literal occurrences, with no within-group edges and cross-group edges exactly for noncomplementary pairs. Draw the conceptual graph for k=3, find an assignment and clique, prove both implications, and bound construction time. This is a research-level exercise.

## 34-E04 Clique and vertex cover in both directions

Let G have vertices {1,2,3}, edges {12,23}, and k=2. Show a clique, an independent set in the complement, and its complementary cover there. For every loopless undirected n-vertex G and $0 \leq k \leq n$, prove CLIQUE(G,k) iff VERTEX-COVER(Gbar,n-k), and give both reductions. This is an advanced exercise.

## 34-E05 Hamiltonian cycle to TSP decision

For a triangle HC instance, construct the complete weighted graph and threshold B. Prove both directions for any n-vertex input. Distinguish a decision certificate from an optimization output. This is a research-level exercise.

## 34-E06 Transitivity and reduction direction

Given $A \leq_p B$ via f and $B \leq_p C$ via g, prove transitivity in full. If C is NP-hard, explain what $C \leq_p L$ implies and why $L \leq_p C$ alone cannot establish hardness of L. This is a research-level exercise.

# NP-completeness exercises (continued)

> Complete these problems before consulting the separate solutions PDF.

## 34-E07 Gate constraints need both implications

For $z = (a \vee b) \wedge \neg a$, use $u = a \vee b, v = \neg a, z = u \wedge v$ to write every gate's CNF and require z=true. Give satisfying wire values. If AND retains only $(z \Rightarrow u) \wedge (z \Rightarrow v)$, omitting $(u \wedge v) \Rightarrow z$, does forcing z=true immediately create a false acceptance in this example? Construct another circuit that connects that AND to a NOT gate and proves the omission is unsound in general.

## 34-E08 Local and global Hamiltonian-cycle gadgets

For the twelve-vertex gadget, let x indicate a3-a4 and y indicate b3-b4. From internal degree two, derive all four cross-edge indicators and external degrees of the ports. List the two cycles when x=y=0 and explain why selectors make this impossible in a global HC. For original path u-v-w and k=1, list the entire 25-vertex cycle with gadget prefixes and check every edge.

## 34-E09 No-carry reduction and reverse decoding

For $F = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee x_3)$, build every Ti,Fi, slack number, and target in column order x1,x2,x3,C1,C2. Use assignment (false,true,true) to reach the target. Show no slack choice rescues the first clause for (false,true,false). Prove no carry for every subset and bound total binary output length.

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https://github.com/xinhi6716/computer-science-self-study

### References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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